

PROJECT REPORT

AI & ML Internship Task 2

Project Title:
**DATA CLEANING & MISSING VALUE HANDLING
USING TITANIC DATASET**

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3. Introduction

3.1 What is Data Cleaning

Data cleaning, also known as data cleansing or data scrubbing, is the critical first step in the data preparation pipeline. It involves identifying and correcting corrupt, inaccurate, incomplete, or irrelevant records from a dataset, table, or database. The purpose of data cleaning is to modify or remove these anomalies to create a uniform, reliable dataset. High-quality data is accurate, consistent, valid, and complete. Cleaning forces the data to meet these rigorous standards.

3.2 Importance of Missing Value Handling

Handling missing values is paramount because most machine learning algorithms operate on matrices and linear algebra; they mathematically fail if fed NaN (Not a Number) values. A clean dataset ensures that feature weights determined during model training reflect reality, leading to stable, trustworthy, and reliable predictions.

3.3 Types of Missing Data

A. MCAR (Missing Completely At Random): The probability of data being missing is the same for all observations. There is no relationship between the missing data and any other values.

B. MAR (Missing At Random): The probability of data being missing is related to some other observed data but not the missing data itself.

C. MNAR (Missing Not At Random): The probability of data being missing is directly related to the value that is missing. (e.g., Passengers who died might be less likely to have certain personal details recorded compared to survivors).

3.4 Applications in Machine Learning

In real-world machine learning applications, algorithms require pristine numerical matrices. Proper data cleaning ensures the structural integrity of predictive models across all domains, from financial forecasting to autonomous driving, preventing catastrophic errors born from faulty assumptions.

4. Problem Statement

Raw datasets acquired from real-world sources contain a mixture of numerical, categorical, and fundamentally flawed data points, most notably missing values. Missing values severely degrade the statistical power of a dataset and physically prevent the execution of standard mathematical algorithms used in predictive modeling. The objective of this project is to process the historical "Titanic Dataset", diagnose its structural voids, and systematically apply targeted statistical imputation and feature deletion techniques to extract a mathematically sound, clean dataset.

5. Project Objectives

1. Detect missing values across all dataset features.
2. Visualize missing data patterns to understand structural flaws.
3. Apply appropriate statistical imputation techniques (Mean, Median, Mode) correctly assigning values back to columns.
4. Remove highly incomplete features that cannot be statistically salvaged.
5. Generate a structurally validated, clean dataset.
6. Prepare the dataset specifically for ingestion by machine learning algorithms.

6. Dataset Description

6.1 Dataset Overview

Dataset: Titanic Dataset

Source: <https://www.kaggle.com/datasets/yasserh/titanic-dataset>

Description: The dataset comprises 891 historical passenger records consisting of 12 diverse attributes covering demographics, socioeconomic status, and survival outcomes.

6.2 Feature Description Table

Feature	Data Type	Description & Purpose
PassengerId	Integer	Unique ID assigned to each passenger.
Survived	Integer	Target Variable (0 = No, 1 = Yes).
Pclass	Integer	Passenger Class (Proxy for socio-economic status).
Name	String	Full Name of the passenger.
Sex	String	Gender of the passenger.
Age	Float	Passenger Age in years.
SibSp	Integer	Count of Siblings/Spouses aboard.
Parch	Integer	Count of Parents/Children aboard.
Ticket	String	Alphanumeric Ticket Number.
Fare	Float	Ticket Fare paid.
Cabin	String	Cabin Number assigned.
Embarked	String	Port of Boarding (C, Q, S).

7. Tools and Technologies Used

- Python 3
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- GitHub
- Microsoft Word

8. Methodology

8.1 Dataset Loading

The Titanic dataset CSV file is ingested into a working environment using Pandas `pd.read_csv()`, creating an active DataFrame object for manipulation.

8.2 Missing Value Detection

The `df.isnull().sum()` function is utilized to generate a precise numerical count of NaN values present vertically within every specific column.

8.3 Heatmap Visualization

A Seaborn heatmap (`sns.heatmap()`) is generated passing the Boolean matrix of missing states to visually map out the exact distribution, sparsity, and clusters of missing data points.

8.4 Missing Value Treatment

To safely update the DataFrame without triggering Pandas `FutureWarning`, explicit column assignment is utilized:

- **Median Imputation (Age):** Extreme outliers are mitigated by filling voids with the median value.
- **Mean Imputation (Fare):** The continuous financial array is balanced by filling gaps with the calculated mathematical average.
- **Mode Imputation (Embarked):** Missing text-based categorical strings are replaced by the most frequently occurring boarding port.
- **Cabin Removal:** Because the column holds an excessive percentage of missing data, it is entirely deleted using `.drop()` to prevent bias.

8.5 Verification

The dataset is re-scanned using `.isnull().sum()` to confirm that the imputation and deletion operations successfully reduced all null counts to zero.

8.6 Dataset Export

The finalized, verified DataFrame is written back to disk as a new CSV file (`Cleaned_Titanic_Dataset.csv`).

9. Implementation and Results

9.1 Python Source Code Implementation

The following Python code was executed to perform the data cleaning, incorporating direct assignment syntax to ensure compatibility with modern Pandas standards:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv('/content/Titanic-Dataset.csv')

# Display first 5 rows
print(df.head())

# Check missing values
print(df.isnull().sum())

# Heatmap for missing values
plt.figure(figsize=(10,5))
sns.heatmap(df.isnull(), cbar=False, cmap='viridis')
plt.title("Missing Values Heatmap")
plt.show()

# Fill missing Age values using Median
df['Age'] = df['Age'].fillna(df['Age'].median())

# Fill missing Fare values using Mean
df['Fare'] = df['Fare'].fillna(df['Fare'].mean())

# Fill missing Embarked values using Mode
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])

# Drop Cabin column
df.drop('Cabin', axis=1, inplace=True)

# Verify missing values after cleaning
print(df.isnull().sum())

# Save cleaned dataset
df.to_csv('Cleaned_Titanic_Dataset.csv', index=False)

print("Dataset cleaned successfully")
```

Figure 9.2: Dataset Preview

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Purpose of Visualization: To observe dataset structure.

Observation: The output reveals columns containing mixed data types (integers, floats, strings) and visible NaN values in the Cabin column.

Interpretation: The dataset contains diverse demographic and numerical characteristics requiring varied processing approaches.

Business/Data Science Insight: Understanding data structure is the first step of preprocessing, dictating how mathematical operations will be applied.

Figure 9.3: Missing Value Count

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

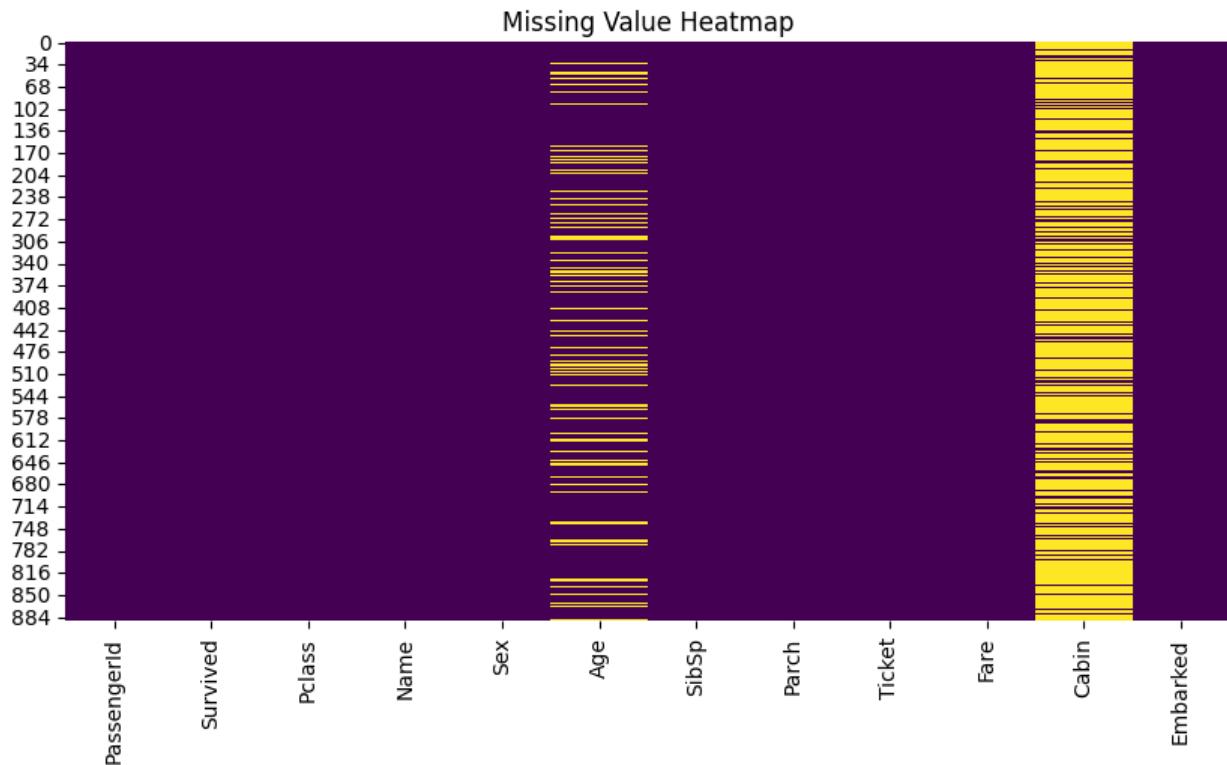
Purpose of Visualization: To identify incomplete features.

Observation: Age (177), Cabin (687), and Embarked (2) contain missing values.

Interpretation: Cabin has the highest missingness, followed by Age, while Embarked is nearly complete.

Business/Data Science Insight: Helps determine a suitable cleaning strategy tailored specifically to the severity of the deficit in each column.

Figure 9.4: Missing Value Heatmap



Purpose of Visualization: To visualize missing value distribution across all features.

Observation: Cabin contains extensive missing values. Age contains moderate missing values. Embarked contains very few missing values.

Interpretation: Missing values are concentrated in specific columns rather than spread randomly, indicating structural issues with the data collection regarding certain variables.

Business/Data Science Insight: Visualization supports informed preprocessing decisions and feature selection. It proves that attempting to impute the Cabin column would fabricate the majority of the data.

Figure 9.5: Missing Values After Cleaning

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Embarked	0

dtype: int64

Purpose of Visualization: To verify successful preprocessing.

Observation: All columns show exactly 0 missing values.

Interpretation: Cleaning procedures successfully resolved data quality issues without destroying the underlying sample size.

Business/Data Science Insight: Clean datasets improve model reliability and prevent computational pipeline failures.

Figure 9.6: Dataset Export Output

	A	B	C	D	E	F	G	H	I	J	K
1	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
2	1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S
3	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C
4	3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	S
5	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	S
6	5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	S
7	6	0	3	Moran, Mr. James	male	28	0	0	330877	8.4583	Q
8	7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463	51.8625	S
9	8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	349909	21.075	S
10	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347742	11.1333	S
11	10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14	1	0	237736	30.0708	C
12	11	1	3	Sandstrom, Miss. Marguerite Rut	female	4	1	1	PP 9549	16.7	S
13	12	1	1	Bonnell, Miss. Elizabeth	female	58	0	0	113783	26.55	S
14	13	0	3	Saunderscock, Mr. William Henry	male	20	0	0	A/5. 2151	8.05	S
15	14	0	3	Andersson, Mr. Anders Johan	male	39	1	5	347082	31.275	S
16	15	0	3	Vestrom, Miss. Hulda Amanda Adolfina	female	14	0	0	350406	7.8542	S
17	16	1	2	Hewlett, Mrs. (Mary D Kingcome)	female	55	0	0	248706	16	S
18	17	0	3	Rice, Master. Eugene	male	2	4	1	382652	29.125	Q
19	18	1	2	Williams, Mr. Charles Eugene	male	28	0	0	244373	13	S
20	19	0	3	Vander Planke, Mrs. Julius (Emelia Maria Vandemoortele)	female	31	1	0	345763	18	S
21	20	1	3	Maselmani, Mrs. Fatima	female	28	0	0	2649	7.225	C
22	21	0	2	Fynney, Mr. Joseph J	male	35	0	0	239865	26	S
23	22	1	2	Beesley, Mr. Lawrence	male	34	0	0	248698	13	S
24	23	1	3	McGowan, Miss. Anna "Annie"	female	15	0	0	330923	8.0292	Q
25	24	1	1	Sloper, Mr. William Thompson	male	28	0	0	113788	35.5	S

Purpose of Visualization: To verify successful export.

Observation: Cleaned dataset generated successfully.

Interpretation: Dataset ready for machine learning deployment.

Business/Data Science Insight: Preprocessed data reduces training errors and establishes a standardized foundation for predictive analytics.

10. Challenges Faced

1. Identifying and resolving Pandas 'FutureWarning' issues related to chained assignments.
2. Handling missing values in Age without skewing the distribution.
3. Deciding mathematically whether the Cabin feature should be statistically imputed or structurally removed.
4. Selecting suitable imputation methods (Mean, Median, Mode) tailored to specific data types.
5. Maintaining overall dataset integrity and record count.

11. Learning Outcomes

1. Mastered Pandas best practices, including direct DataFrame assignment overrides to prevent deprecation errors.
2. Learned robust data cleaning techniques suitable for complex datasets.
3. Learned in-depth missing value analysis and classification.
4. Learned visual heatmap generation and structural interpretation.
5. Learned the complete dataset preprocessing workflow required for professional AI engineering.

12. Conclusion

The data cleaning phase applied to the Titanic dataset successfully diagnosed and resolved critical data quality issues. Through rigorous missing value detection and subsequent heatmap analysis, structural voids within the data were accurately mapped. By applying logical statistical imputation—median for age, mean for fare, and mode for categorical variables—while adhering to proper Pandas syntax guidelines, the integrity of the numerical distributions was preserved. The strategic removal of the highly compromised Cabin feature prevented the introduction of synthetic noise. Ultimately, the dataset quality improvement was dramatic, resulting in a finalized, robust matrix completely free of NaN values, marking the final dataset's strict readiness for advanced machine learning algorithms.

13. References

1. Titanic Dataset:
<https://www.kaggle.com/datasets/yasserh/titanic-dataset>

2. Google Colab Notebook:

https://colab.research.google.com/drive/1Dzrh7EupG_FMEAyvNVKhzGMj7YNIBifb?usp=sharing

3. GitHub Repository:

<https://github.com/dbmodi2311/Data-Collection-Data-Source-Identification/tree/main/Task%202>

14. Appendix

Appendix A: Complete Python Source Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
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# Drop Cabin column
df.drop('Cabin', axis=1, inplace=True)

# Verify missing values after cleaning
print(df.isnull().sum())

# Save cleaned dataset
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print("Dataset cleaned successfully")
```